Budget Tiers: $15,000 to $50,000

Assistive Systems for Trusted Robotic Operations

R0.4 draft · 2026-11-04 · Owner: Faisal Farouq Status: PLANNING ONLY. Only the items marked *listing* are priced from today's web listings. Everything else is an estimate, not a quote. Nothing is ordered.

The short version

- The core walking robot costs about $19,550 if you pay for everything except the test space (range $13,968–$25,139). So $15k works only if your contacts lend the gantry, the test space, the reviews and the tools.
- Recommended target: $25k. That's the core robot, paid for, plus the spares and sensors that keep walking tests going when something breaks, with a 12% reserve.
- Above $25k, money mostly buys personality and comfort, not a higher chance of walking. Arms, a moving head and a nicer shell come at $35k; motion capture and a spare leg at $50k.
- R0.1 under-budgeted the frame. It allowed $650 for structure; a realistic figure is $1,500–3,000.
- Money you need in the next 60 days: about $2,698–$5,149 (bench actuator, CAN adapters, power supply, stop chain, bench rig, one-leg rig, computer, IMU, wiring, care-demo hardware, tools).
- Availability risk: one seller lists the RobStride 04 as sold out. Ask for a written quote for 11 units (17 at $50k) now.

Tiers at a glance

Totals use the middle of each estimate range; the reserve is what's left of the tier. Each tier includes everything in the tiers above it.

Tier	What you get	Mid cost	Reserve	Must come from your contacts
$15k · Minimum walker	Core robot that can take tethered steps on cable power. No battery, arms pinned, printed shell. Only works if your contacts cover the items on the right.	$13,878	$1,122 (7%)	Overhead gantry or an approved beam, test space, reviews, tools, 3D-printer access
$20k · Core, paid for	Everything in $15k, plus paid reviews, your own tools, and a battery with safe charging.	$17,554	$2,446 (12%)	Gantry or approved beam, test space, 3D-printer access
$25k · Reliable core (recommended)	Adds your own gantry, a second spare actuator, a calibrated IMU, a second power supply and a CAN analyser. The cheapest level where one broken part or a noisy sensor doesn't stop walking.	$21,960	$3,040 (12%)	Test space, 3D-printer access
$35k · Expressive ASTRO	Adds moving arms (2 joints each) and a head that turns, machined gear stages, a shop-made shell, a second battery, a wireless e-stop and more review time. It starts to *feel* like ASTRO.	$30,004	$4,996 (14%)	Test space, 3D-printer access

Tier	What you get	Mid cost	Reserve	Must come from your contacts
$50k · Research grade	Adds motion capture (true body position for tuning the gait), a complete spare leg, a machined frame and a rented test space.	$42,849	$7,151 (14%)	3D-printer access
Beyond $50k	Foot force/torque sensors, 4-joint arms, composite shell, full independent safety review, a bigger computer, a brand-name gantry.	≈$68,644	—	—

If a contact can't provide something, add it from the next tier: gantry $1,500–2,500, paid reviews $1,500–3,000, tools $300–700, test space $1,000–2,500 (all estimates).

Parts by tier

$15k · Minimum walker

These are the parts every tier needs. Running total $10,118–$17,639.

Part	Specification	Cost	Price basis
10 × Leg actuators	RobStride 04, 120 N m peak / 40 N m rated, 48 V, 1.42 kg	$3,190	listing
Bench + spare actuator	RobStride 04; used on the single-joint bench first, then kept as the spare	$319	listing
4 × Gear stages	Belt or printed planetary: 3:1 on both hip rolls, 2:1 on both knees	$200–$600	estimate
2 × USB-CAN adapters	USB to CAN 2.0, 1 Mbit/s (one per leg bus)	$80–$180	estimate
Main computer	Jetson Orin Nano Super 8 GB + NVMe SSD	$249–$330	listing
2 × IMU	Consumer 9-axis IMU board (BNO085 class), 1 spare	$50–$120	estimate
Foot load cells	4 load cells per foot + 2 amplifier boards	$80–$200	estimate
Bench / tether power supply	48 V, ~1 kW enclosed supply (Mean Well RSP-1000-48 class)	$250–$400	estimate
Stop chain	2 wired e-stops, safety relay, 48 V DC contactor, fuses, breaker	$300–$700	estimate
Wiring and connectors	CAN cable, XT60/XT90, DC-DC converters, enclosures	$300–$600	estimate
Frame parts	Laser-cut / CNC aluminium plates and tubes from an online shop	$1,500–$3,000	estimate
Bearings, shafts, fasteners	—	$300–$600	estimate
Feet	Aluminium plates + rubber soles	$100–$250	estimate
3D-printing material	PETG / nylon; assumes printer access	$200–$400	estimate
Single-joint bench	Aluminium extrusion frame, lever arm, weights, load cell	$200–$400	estimate

Part	Specification	Cost	Price basis
One-leg rig	Extrusion frame, vertical linear guide, hanging support	$500–$1,000	estimate
Trolley, tether, rated connectors	Rigger specifies; includes slack adjuster	$300–$700	estimate
Tether load cell	In-line load cell + amplifier	$100–$300	estimate
Chain hoist	Manual chain hoist on the gantry	$150–$350	estimate
Shell panels	3D-printed / thermoformed panels and mounts	$600–$1,500	estimate
Handling stand	Guided linear actuator ≥ 98 N holding, controller, limit switches, base (D-04)	$400–$900	estimate
Voice and screen	USB microphone/speaker, 10 in touchscreen	$150–$400	estimate
Shipping, tariffs, tax	~15% on imported parts (actuators, compute, sensors)	$600–$1,200	estimate

$20k · Core, paid for

Added to the tier above: $2,350–$5,000. Running total $12,468–$22,639.

Part	Specification	Cost	Price basis
Robot battery	48 V nominal, 15-20 Ah, with BMS	$400–$900	estimate
Charger + fire-safe charging	Matched charger, metal charging container, Class ABC extinguisher	$150–$400	estimate
Tools if not owned	Multimeter, torque wrench, crimpers, USB oscilloscope	$300–$700	estimate
Paid reviews	Limited hours: structural/mechanical, rigging, battery/electrical	$1,500–$3,000	estimate

$25k · Reliable core (recommended)

Added to the tier above: $3,569–$5,244. Running total $16,037–$27,883.

Part	Specification	Cost	Price basis
Overhead gantry	1-ton adjustable aluminium gantry, ≥ 10 ft under beam (budget brand)	$1,500–$2,500	estimate
Second spare actuator	RobStride 04	$319	listing
Better IMU	VectorNav VN-100 Rugged development kit	$1,200–$1,525	listing
Second power supply	Same 48 V supply, for parallel or spare	$250–$400	estimate
CAN analyser	PCAN-USB class interface + software	$300–$500	estimate

$35k · Expressive ASTRO

Added to the tier above: $5,594–$10,494. Running total $21,631–$38,377.

Part	Specification	Cost	Price basis
Machined gear stages	Upgrade printed/belt stages to machined planetary (extra cost over tier 1)	$800–$1,800	estimate

Part	Specification	Cost	Price basis
Wireless e-stop	Rated wireless stop for the stop operator	$400–$1,000	estimate
Second battery	Same pack as tier 1	$400–$900	estimate
Extra review hours	More reviewer time before G2 and G4	$1,000–$2,000	estimate
4 × Moving arms (2 DoF each)	RobStride 03 (60 N m peak), shoulder pitch + elbow, both arms	$896	listing
Arm brackets and covers	—	$300–$600	estimate
2 × Head pan/tilt	RobStride 05 (5.5 N m)	$298	listing
Better shell	Shop vacuum-formed panels (extra cost over tier 1)	$1,500–$3,000	estimate

$50k · Research grade

Added to the tier above: $10,595–$15,095. Running total $32,226–$53,472.

Part	Specification	Cost	Price basis
Machined frame upgrade	Shop-machined joints and brackets (extra cost over tier 1)	$2,000–$4,000	estimate
Test-space rental	If no free space is approved	$1,000–$2,500	estimate
Motion capture	OptiTrack Flex 3, 6-camera bundle	$6,000–$7,000	listing
5 × Spare leg set	5 x RobStride 04	$1,595	listing

Beyond $50k

Added to the tier above: $17,595–$33,995. Running total $49,821–$87,467.

Part	Specification	Cost	Price basis
Rated brand-name gantry	RAZE 1-ton, 12 ft span, 8-12 ft (replaces budget gantry; extra cost)	$2,700–$3,700	listing
Bigger computer	Jetson AGX Orin 64 GB developer kit	$1,999–$5,399	listing
2 × Foot force/torque sensors	6-axis F/T sensor per foot (Bota SensONE class)	$6,000–$12,000	quote needed
4 × Full arms (4 DoF each)	+2 RobStride 03 per arm (shoulder roll, wrist)	$896	listing
Composite shell	Professional fibreglass / composite panels (extra cost over tier 3)	$3,000–$6,000	estimate
Full independent safety review	Documented review of the complete walking setup	$3,000–$6,000	estimate

Things to know before ordering

- Mass: the $35k and $50k tiers add arm and head actuators. Re-run the mass budget before ordering them; the 55 kg cap still applies.
- Arms are for gestures only. They never lift, catch or hold a person (ASTRO-R-059).
- Battery waits: Year 1 walking is tethered, so the $15k tier runs on cable power. Buy the battery only after the power tests (about Day 220).

- Tariffs and shipping: about 15% is included for imported parts. Check the rate when you order.
- Funding timing: you said you can raise money when parts are ordered. The first actuator, power supply and stop chain are needed around Day 20. Raise that first $2,700–5,100 now (the 60-day list above).

For comparison: buying a walking robot

Unitree's H2 (182 cm) base model is $29,900 from Unitree or $40,900 from a US store, but the base model has no developer access. The programmable EDU version starts at $66,800. A full-size walker you can program yourself costs more than this whole $50k range, so building is the cheaper path.

Sources (checked 2026-11-04)

- RobStride listings: rcdrone.top RobStride collection; openelab RobStride 04 (EUR 254.75, sold out; specs 40/120 N m, 1.42 kg, 48 V)
- Jetson Orin Nano Super, USD 249
- VectorNav VN-100 Rugged Development Kit, USD 1,200–1,525
- OptiTrack Flex 3: 6-camera bundle under USD 7,000
- Crane Depot RAZE 1-ton gantry, 12 ft span, USD 5,200
- Jetson AGX Orin 64 GB listing, USD 5,399 (sold out); NVIDIA launch list USD 1,999
- Humanoid prices, roboticscenter.ai, updated 2026-10-18
- CubeMars AK10-9 V3.0, USD 922.99 (18/53 N m): checked as an alternative; the RobStride 04 gives more torque for a third of the price
- R0.1 budget/financial_budget.csv for the old allowances

Full line list: budget/budget_tiers_R0_4.csv.